



Nishant Bhawe

Fourth Year Undergraduate
Electrical Engineering Dual Degree
Indian Institute of Technology Bombay

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University	Degree	Year	CPI
IIT Bombay	B.Tech Electrical Engineering, M.Tech Controls and Computing	2022–27	9.12

Pursuing a **Dual Minor** in **Robotics** and **Machine Learning (C-MInDS)** at IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Recipient of **Undergraduate Research Award** for outstanding work on Bilevel Optimization [2025]
- Secured **All India Rank 964** in the JEE Advanced out of **0.16 million+** candidates [2022]
- Qualified for **scholarship** and acquired **AIR 262** in Kishore Vaigyanik Protsahan Yojana [2022]
- Secured a **Branch Change** (top **9%** of **1400+** students) to Electrical Engineering Department [2023]
- Received **Silver Medal** in Dr. Homi Bhabha Balvaidnyanik Science Talent Search Competition [2020]

RESEARCH EXPERIENCE

Network-Optimal Decentralized Bilevel Optimization *Under Review* [May'26 - Present]
Guide: Prof. Mayank Baranwal, Indian Institute of Technology Bombay

- Developed a **single-loop** decentralized bilevel optimizer combining **Chebyshev-accelerated consensus** and **optimistic gradient tracking** for efficient networked optimization.
- Established **communication lower bounds** and **network-optimal complexity**, isolating the independent effects of network, outer-level, and inner-adjoint conditioning.

Hybrid Reach-Avoid MPC with Backstepping Control *Under Review* [Oct'25 - Dec'25]
Guide: Prof. Shankar Deka, Aalto University

- Designed a hybrid **Reach-Avoid MPC** backstepping controller to address **infeasibility** issues of **finite horizon MPC** by integrating Exponential Control Guidance-Barrier Functions for safety and reachability.
- The optimizer's feasibility is ensured using the guarantees provided by the reach-avoid set from ECGBF.

Vision-based Unmanned Aerial Manipulator Control [May'25 – Jul'25]
Guide: Prof. Lin Zhao, CORE Lab, National University of Singapore

- Evaluated pose estimation frameworks and deployed **FoundationPose** for real-time 6D object tracking
- Utilized hybrid **MPC + L1 adaptive controller** for trajectory tracking under disturbances during dynamic high-speed pick-up maneuvers, accounting for uncertainties in model parameters.
- Demonstrated vision-based closed-loop grasping of a can in both PX4-Gazebo simulation and **real-world drone experiments** with **YOLO-based online segmentation** for initial mask generation.

Lane-Less Autonomous Driving For Indian Roads *Published at CODIT* [Jan'26 - Present]
Guide: Prof. Parthib Khound and Honda R&D

- Developing a **lane-less autonomous-driving** planning & control pipeline for India roads using kinematic-bicycle vehicle model, sector-based perception, and a PD tracking controller.
- Implemented a **repulsive-potential lookahead planner** to replace heuristic lane-centering and improve worst-case clearance decisions.
- Built a **CARLA** simulation & evaluation framework with randomized traffic scenarios & collision detection

Gradient-Flow Methods for Bilevel Optimization *Under Review* [Jul'25 - Dec '25]
Guide: Prof. Mayank Baranwal, Indian Institute of Technology Bombay

- Applied **PL** inequality to the lower-lever function of the continuous bilevel optimization problem and the **non-convex upper-level** function to guarantee **asymptotic** convergence under **minimal** assumptions
- The approach uses a **hypergradient estimator** with **Tikhonov** regularization to avoid singularity issues in Hessian inverse and **projection** operator to avoid unbounded gradients.

Quantum Computing for Autonomous Driving *Published at IV Symposium* [Dec'24 - Present]
Guide: Prof. Lingxi Li, Purdue University

- Explored quantum computing applications in **motion control**, trajectory planning for autonomous driving
- Developing a **quantum control** algorithm to optimize **motion planning** for autonomous driving
- Exploring **hybrid quantum-classical** approaches to enhance existing control algorithms in dynamic driving environments by using **quantum superposition**

Point Cloud Generation from Raw Data *Under Review* [May'24 - Mar'25]
Guide: Prof. Leena Vachhani

- Engineered **Grid Segmentation** Algorithm for data processing and intelligent selection of vehicles relevant for real-time path planning and **collision avoidance** with **minimal computational complexity**
- Identified the algorithm to be very efficient in a comparative analysis with Ball trees, ANN, and KD trees
- Reduced computation time considerably to **0.193 s/timestamp** using **Grid Segmentation** in python

PROFESSIONAL EXPERIENCE

Robotics R&D Intern | ABB Innovation Centre | Robotics Lab [May'24 - Jul'24]

- Researched, compared, and implemented an optimal two-phase Recursive Partitioning Algorithm in .NET C# for single-SKU palletizing, a variant of **2D Bin Packing** problem
- Developed a mixed SKU pattern generator, **single-handedly** engineering an **algorithm** with 4 variants to ensure gripper safety margins, account for placement errors, and **enhance stability** during transportation
- Implemented the **recursive algorithm** with **backtracking** for feasibility checks and robust performance

KEY PROJECTS

Fault-Tolerant Controller for Single Rotor Failure | Inter-IIT Tech Meet [Oct'24 - Dec'24]

- Secured **2nd** position for developing a fault-tolerant controller for a quadrotor under **single-rotor failure**
- Employed **NDI** combined with **feedback linearization** for quick response to motor failure and stabilization
- Implemented a modified **Two-Stage Kalman Filter** for real-time failure detection using only the onboard IMU and GPS. It generates a control-effectiveness matrix, which adapts to motor failure within **0.03s**.
- **Validated** control and fault-detection algorithms in Gazebo simulation as well as **hardware tests**, relying solely on the PX4 flight controller stack, with no external computation devices or additional sensors.

Thanopod | Team genSUS, IIT Bombay | Robofest Gujarat 4.0, GUJCOST [Jun'24 - Jan'25]

- Conducted a **literature review** on the control of self-balancing bots and autonomous navigation
- Simulated motor dynamics for **Thanopod 2-Wheeled Self-Balancing Bot** to model the non-linear system for designing **LQR** Controller for enhanced stability, estimation and precise motion control
- Designed a **cascaded PID** loop ensuring stable balance with smooth motion along a **curvilinear** track
- Applied advanced **computer vision** and **stanley controller** for effective **line detection** and **tracking**

Pulse Compression Using Barker Codes [Jan'24 - Apr'24]
Guide: Prof. Kushal Tuckley and Indian Army

- Implemented a **radar signal processing** system using **Barker codes** for pulse compression, achieving **7x** improvement in range resolution in detecting transmitted signals under noisy conditions.
- Designed and simulated a complete **communication pipeline** in MATLAB, including Barker code modulation, AWGN channel modeling, matched filtering, and binary message recovery.

TECHNICAL SKILLS

Languages & Libraries	Python, MATLAB, ROS, Linux, C, C++, C#
Softwares	PX4, MATLAB, Simulink, Gazebo, ArduinoIDE, Eagle, L ^A T _E X

KEY COURSES

Robotics	Guidance and Control of UAVs, Control systems, Adaptive & Learning Control, Multivariable Control, Optimal Control, Computer Vision
Math & CS	Applied Linear Algebra, Linear Algebra Differential Equations, RL, Optimization, Nonlinear Systems, Intro to DS and ML, Distributed Optimization and ML
Electrical	Microprocessors, Digital Systems, Analog Circuits, Probability , Signal Processing